

Model Selection

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Texas Suffers From A Doctor Shortage

By JONATHAN BAKER • NOV 1, 2017



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When it comes to having a high ratio of doctors to citizens, the State of Texas ranks near the bottom. In fact, [as The Dallas Morning News reports](#), 43 states have a higher proportion of primary care physicians to residents than Texas.

And West Texas suffers from a lack of doctors more than other parts of the state. There are 80 counties in Texas with five or fewer practicing doctors - many in West Texas. Thirty-five Texas counties have [no doctors at all](#).





- Data from all counties in Texas with 10,000+ residents
- Let's try to model (explain) what factors predict the number of physicians per 10,000 people in a county
- What factors do you think would cause a county to have more (or fewer) physicians?



Potential predictor variables

- **LandArea**: Area in square miles
- **PctRural**: Percentage rural land
- **MedianIncome**: Median household income
- **Population**: Population
- **PctUnder18**: Percent children
- **PctOver65**: Percent seniors
- **PctPoverty**: Percent below the poverty line
- **PctUninsured**: Percent without health insurance
- **PctSomeCollege**: Percent with some higher education
- **PctUnemployed**: Percent unemployed

How do we know which model is best?

Overfitting

Model selection metrics

Best subsets regression

Backward stepwise regression

Training and test sets



Metrics for measuring model fit: R^2

- R^2 measures the proportion of variance in Y that is explained by the predictors
- It never decreases when you add more predictor variables
- So we can't just pick the model that maximizes R^2 — would cause us to always include every variable!

Suppose we are trying to predict the % of the vote the Democratic candidate for US president will get in the 2024 election...

THE
PROBLEM
WITH STATEMENTS LIKE

"NO <PARTY> CANDIDATE HAS
WON THE ELECTION WITHOUT <STATE>"

OR

"NO PRESIDENT HAS BEEN
REELECTED UNDER <CIRCUMSTANCES>"

★UPDATED FOR 2020★

1788...

NO ONE HAS BEEN
ELECTED PRESIDENT
BEFORE.



...BUT WASHINGTON WAS.

1792...

NO INCUMBENT HAS
EVER BEEN REELECTED.



...UNTIL WASHINGTON.

1796...

NO ONE WITHOUT
FALSE TEETH HAS
BECOME PRESIDENT.



...BUT ADAMS DID.

1800...

NO CHALLENGER HAS
BEATEN AN INCUMBENT.



...BUT JEFFERSON DID.

1804...

NO INCUMBENT HAS
BEATEN A CHALLENGER.



...UNTIL JEFFERSON.

1808...

NO CONGRESSMAN HAS
EVER BECOME PRESIDENT.



...UNTIL MADISON.

1812...

NO ONE CAN WIN
WITHOUT NEW YORK.



...BUT MADISON DID.

1816...

NO CANDIDATE WHO
DOESN'T WEAR A WIG
CAN GET ELECTED.



...UNTIL MONROE WAS.

1820...

NO ONE WHO WEARS
PANTS INSTEAD OF
BREECHES CAN BE
REELECTED.



...BUT MONROE WAS.

1824...

NO ONE HAS EVER
WON WITHOUT A
POPULAR MAJORITY.



...J.Q. ADAMS DID.

1828...

ONLY PEOPLE FROM
MASSACHUSETTS AND
VIRGINIA CAN WIN.



...UNTIL JACKSON DID.

1832...

THE ONLY PRESIDENTS
WHO GET REELECTED
ARE VIRGINIANS.



...UNTIL JACKSON.

1836...

NEW YORKERS
ALWAYS LOSE.



...UNTIL VAN BUREN.

1840...

NO ONE OVER 65 HAS
WON THE PRESIDENCY.



...UNTIL HARRISON DID.

1844...

NO ONE WHO'S LOST HIS
HOME STATE HAS WON.



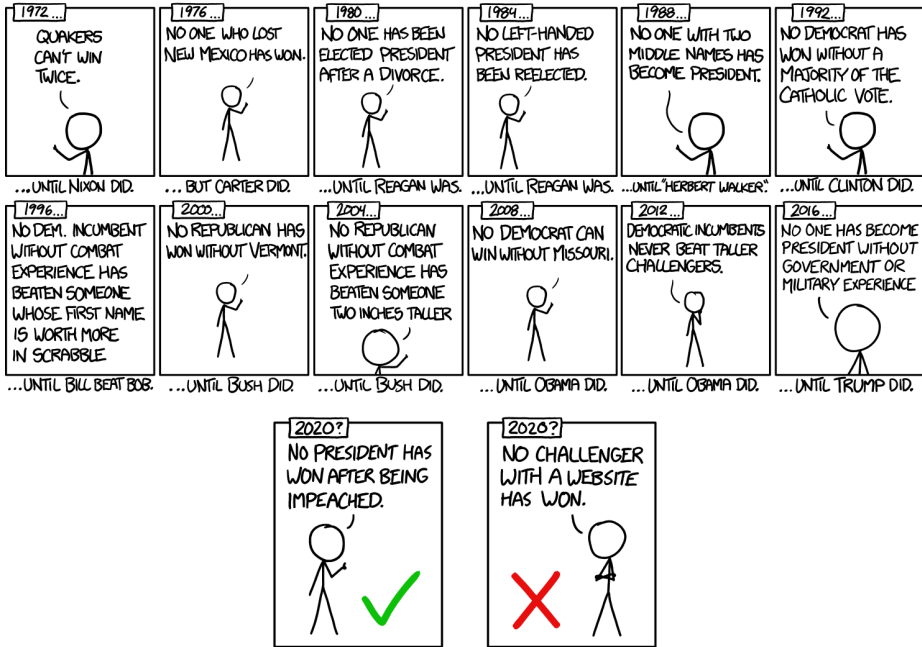
...BUT POLK DID.

1848...

AS GOES MISSISSIPPI,
SO GOES THE NATION.



...UNTIL 1848.



CONGRATULATIONS TO PRESIDENT-ELECT
JOE BIDEN FOR BREAKING THE WEBSITE CURSE!

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- ...

But how much would you trust a model with dozens of variables like this?



Before we look at another example of overfitting, a definition:

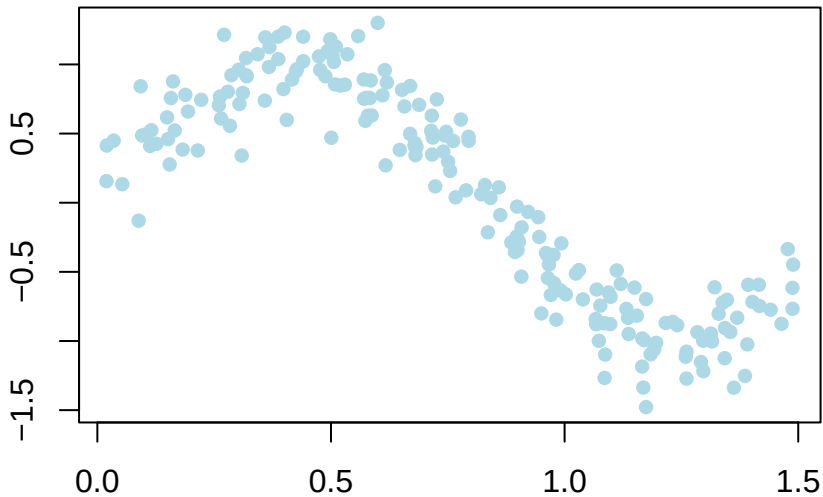
- The mean squared error is just the average of the sum of squared errors (residuals):

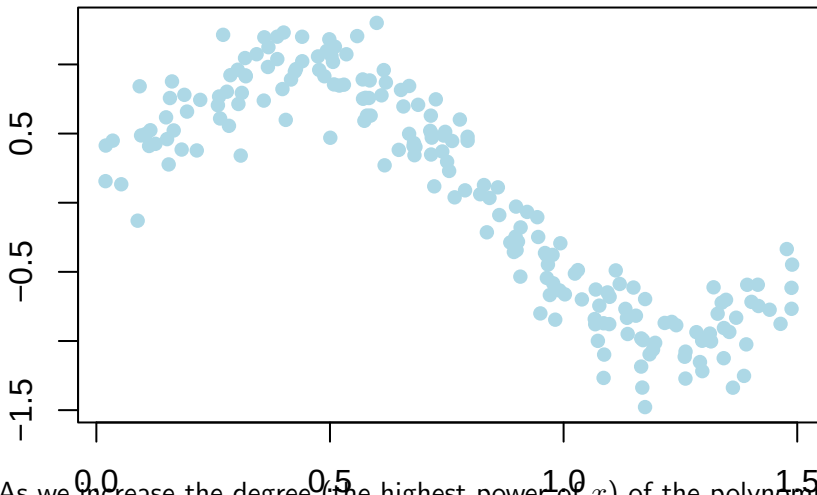
$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- We'll often want to use the root mean squared error (RMSE) because it will be interpretable in the same units as Y :

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

- **Lower** numbers indicate better-fitting models!

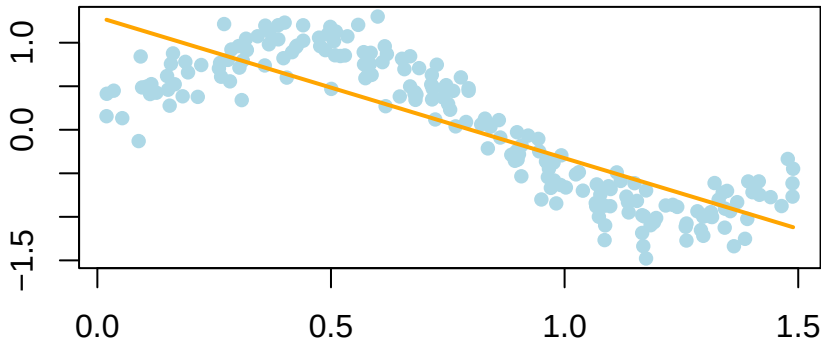




As we increase the degree (the highest power of x) of the polynomial, the fit improves, since the curve can zig and zag to capture more of the idiosyncrasies of the sample.



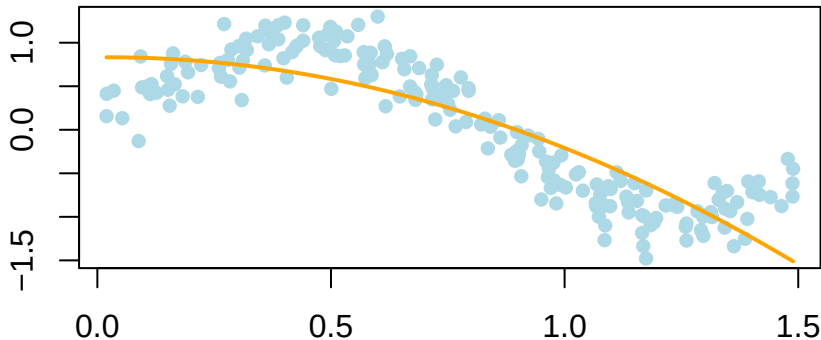
Degree 1 polynomial



RMSE = 0.4152



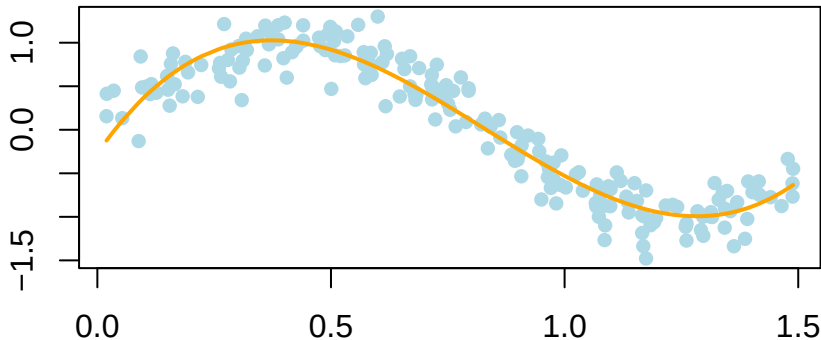
Degree 2 polynomial



RMSE = 0.3801



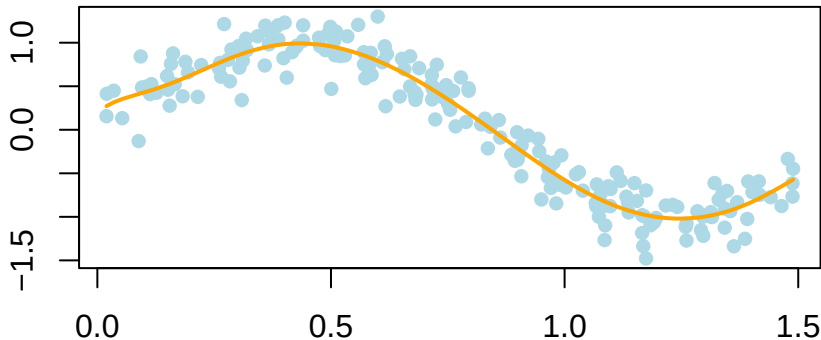
Degree 3 polynomial



RMSE = 0.2159



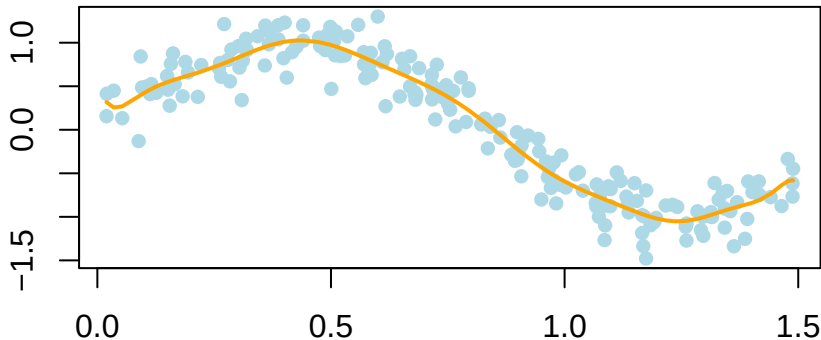
Degree 10 polynomial



RMSE = 0.1978



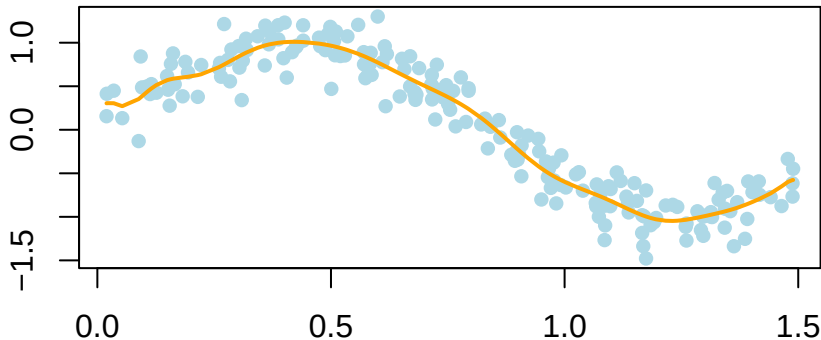
Degree 20 polynomial



RMSE = 0.1965

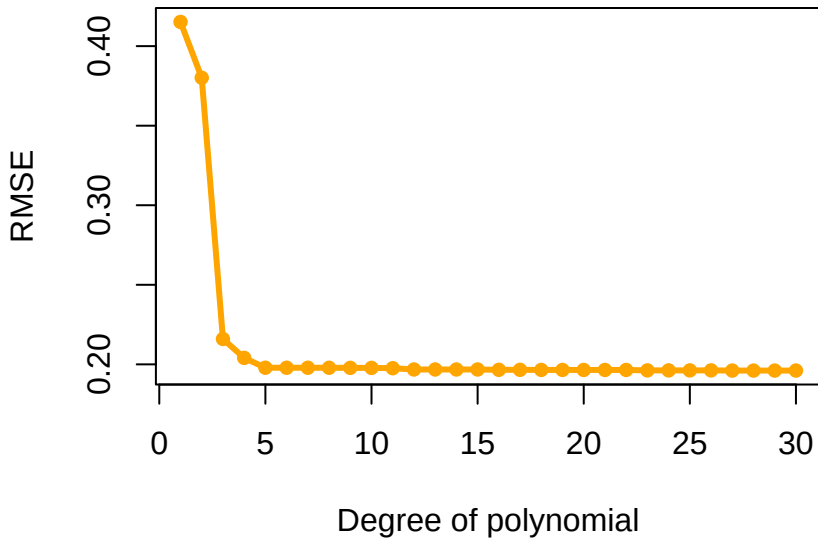


Degree 30 polynomial



RMSE = 0.1961

As we fit increasingly complex polynomials, the MSE decreases:

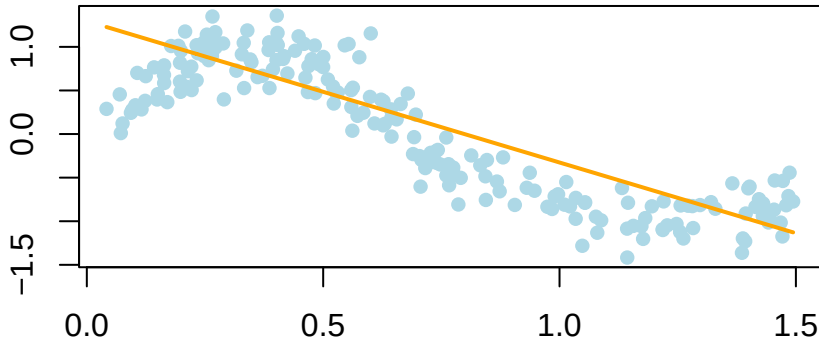




- The best models are **parsimonious**: those that strike a balance between fitting well and being simple.
- Besides being easier to communicate, there's another reason to prefer simpler models—they tend to generalize better to new data.
- Let's see how these models perform on a new sample from the same population!



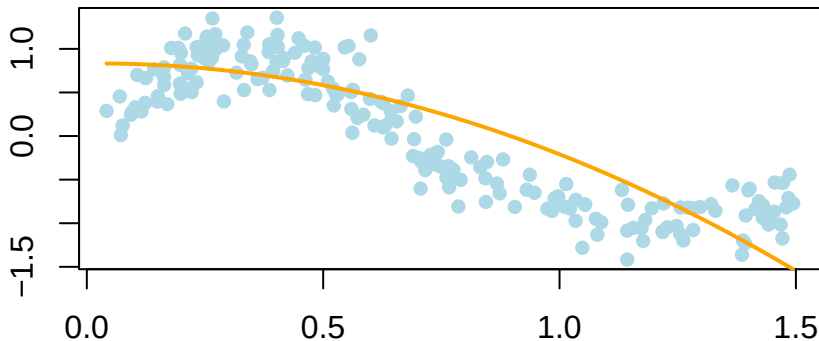
Degree 1 polynomial



RMSE = 0.4083 (was 0.4152 in the original sample)



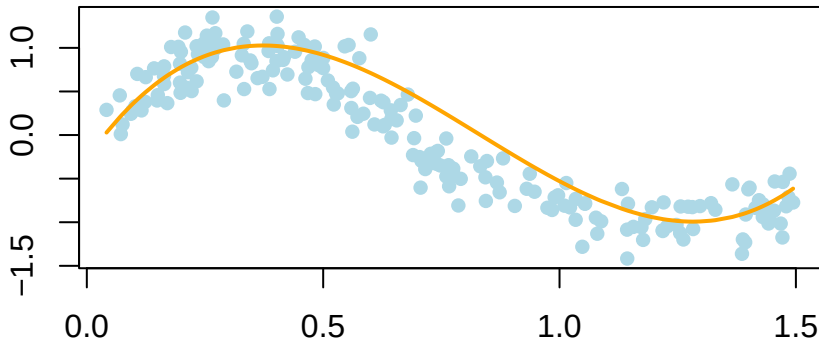
Degree 2 polynomial



RMSE = 0.4401 (was 0.3801 in the original sample)



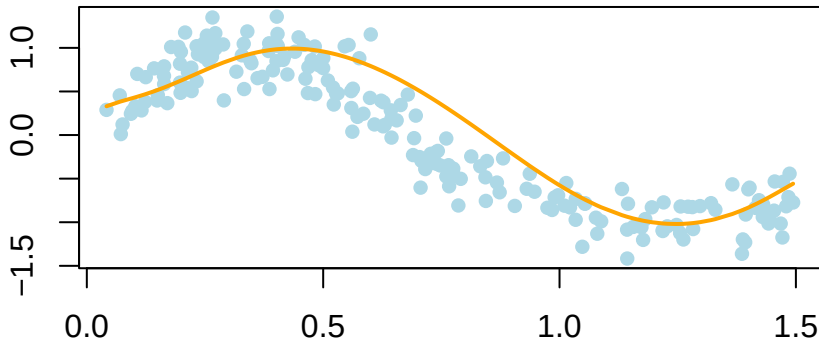
Degree 3 polynomial



RMSE = 0.3395 (was 0.2159 in the original sample)



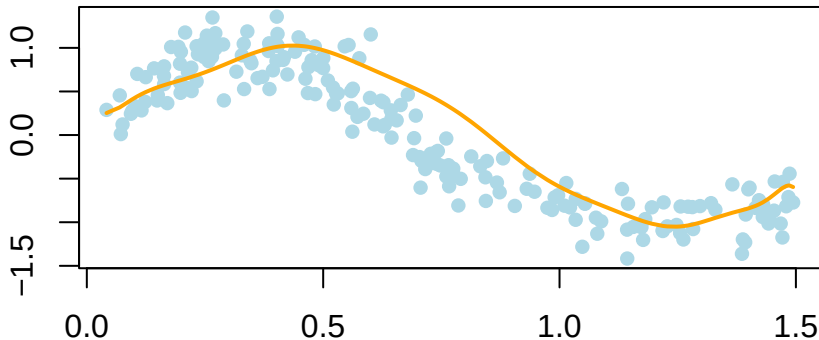
Degree 10 polynomial



RMSE = 0.3693 (was 0.1978 in the original sample)



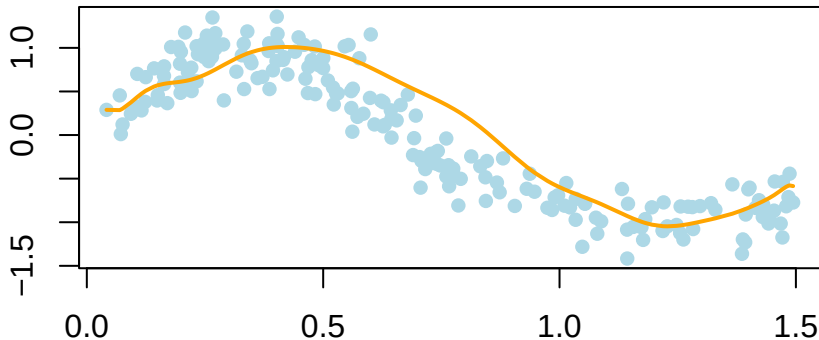
Degree 20 polynomial



RMSE = 0.3725 (was 0.1965 in the original sample)

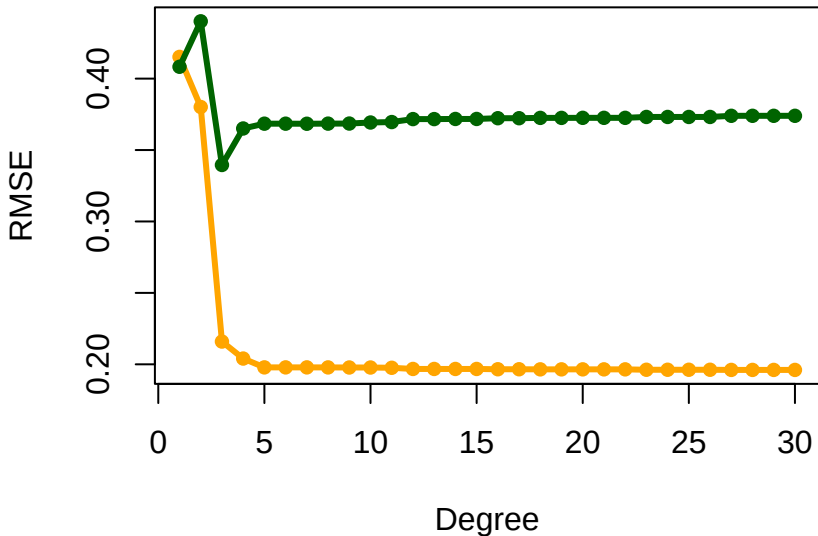


Degree 30 polynomial



RMSE = 0.374 (was 0.1961 in the original sample)

The increasingly complex polynomials do not perform as well on the new data (green) as the simple polynomials, because they had **overfit** the idiosyncrasies of the original data (orange).





Overfitting in multiple regression

- A similar phenomenon occurs when you have many variables to choose from when building a regression model



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- But the more complex models are more likely to be **overfitting** the data, and won't generalize as well as simple models



Overfitting in multiple regression

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- By including many variables in your model, you can get R^2 to increase (or RMSE to decrease) on the data used to build the model
- But the more complex models are more likely to be **overfitting** the data, and won't generalize as well as simple models
- In general, a model's performance on the data used to build the model will usually be stronger than its performance on new data

Overfitting

Model selection metrics

Best subsets regression

Backward stepwise regression

Training and test sets

To avoid overfitting, we can try using a different model fit metric (instead of R^2) that penalizes us for adding more variables.



Metrics for measuring model fit: Adjusted R^2

- Adjusted R^2 applies a penalty to R^2 based on the number of predictors k and number of cases n :

$$\text{Adjusted } R^2 = 1 - (1 - R^2) \frac{n - 1}{n - k - 1}$$

- The idea is that you are penalized more by adding more predictor variables, so adjusted R^2 won't always increase by adding more variables
- Useful because it “almost” has the nice variance-explained interpretation of R^2
- Higher** numbers indicate better-fitting models



Metrics for measuring model fit: AIC

- The AIC (Akaike information criterion) penalizes models with more predictors
- **Formula:** $AIC = 2k - 2 \ln(L)$
 - k = number of parameters (predictors + intercept)
 - L = likelihood of the data given the model
- For linear regression: $AIC = 2k + n \ln(RSS/n)$
 - RSS = residual sum of squares
 - n = sample size
- **Lower** numbers indicate better-fitting models



Intuition behind AIC

AIC balances two competing goals:

1. **Fit the data well** → maximize likelihood (minimize RSS)
 - More predictors $\Rightarrow$ better fit $\Rightarrow$ higher likelihood
2. **Keep the model simple** → penalize complexity
 - More parameters $\Rightarrow$ penalty term $2k$ increases

The tradeoff: $AIC = \text{goodness of fit} - \text{complexity penalty}$

- Adding a predictor improves fit but increases penalty
- AIC picks the model where the improvement in fit is worth the added complexity

There are a *lot* of possible models—how do we find the one with the best (highest) Adjusted R^2 or best (lowest) AIC?

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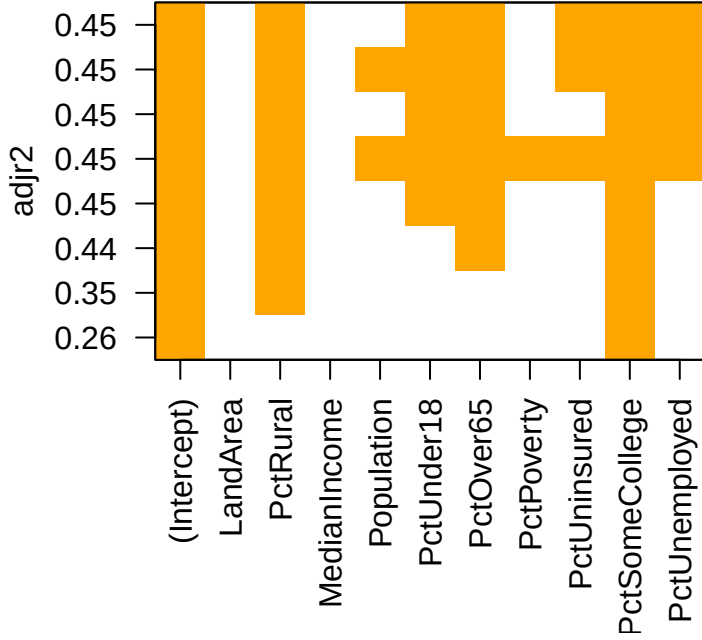
Backward stepwise regression

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Best subsets regression

- Computers are fast! Just let R try out all of the $2^k - 1$ possible models for you (k = number of predictors).
- R will present you the model with the best Adjusted R^2 for each possible number of predictors.



- Best-subsets regression presents us with a candidate model for each possible number of predictors.
- The label on the y -axis shows the Adjusted R^2 value for the model corresponding to the filled-in squares for that row



The downside of best subsets regression

When there are a large number of predictors, even R is too slow to run all possible models!

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1. Start with a “full” model containing all of the predictors.
2. Compare the model's Adjusted R^2 (or AIC) with the Adjusted R^2 (or AIC) of all possible models that result from removing one predictor variable.
3. If removing a predictor results in a better Adjusted R^2 (or AIC), remove that predictor variable and repeat step 2. If we can't improve Adjusted R^2 (or AIC) by removing a predictor variable, stop.



Implementing backward stepwise regression

```
full <- lm(PhysiciansPer10000 ~ LandArea + PctRural +  
          MedianIncome + Population + PctUnder18 +  
          PctOver65 + PctPoverty + PctUninsured +  
          PctSomeCollege + PctUnemployed,  
          data=counties)  
step(full)
```



The downside of stepwise regression

Stepwise regression will not necessarily give you the best model; by only adding or removing one variable at a time, you can get locked into a particular “path” that means you may never consider better models.

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Evaluating model performance

- So far, we've been evaluating a model's performance based on how well it predicts the data we used to fit the model



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Evaluating model performance

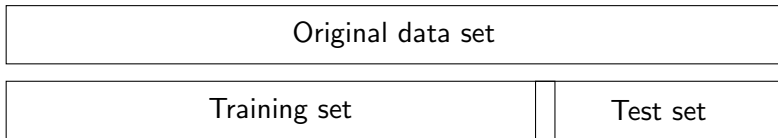
- So far, we've been evaluating a model's performance based on how well it predicts the data we used to fit the model
- This is...kind of cheating! (It's what explains that crazy xkcd comic about presidents.)
- And what we really care about is how well the model will perform on **new** data



- **Overfitting** happens when creating a model that fits your *sample* very well but does not generalize well to the larger *population* (like the presidential election example!).
- In other words, an overfit model is one where the Adjusted R^2 /AIC (or prediction accuracy for logistic regression) is high, but the model will not work as well as expected when given new data.



A solution: training and test sets



- To figure out how much a particular model is overfitting, we can split up our data into a **training set** and a **test set** (a typical split is 70% training set / 30% test set).
- We use the **training set** to build the model, and then evaluate the quality of the model on how well it predicts Y in the **test set**.
- This gives us a better estimate on how well the model will work to predict Y on new data.

Let's split our data into a training and test set with a rough 70/30 split. ($n = 168$, and 30% of 168 is approximately 50.)

```
# Select which row numbers will correspond to test cases  
test.cases <- sample(1:168, 50)  
  
# Build the test set from those row numbers  
test.set <- counties[test.cases,]  
  
# Build the training set from the rest  
training.set <- counties[-test.cases,]
```

Let's build the model using the training set:

```
full <- lm(PhysiciansPer10000 ~ LandArea + PctRural +  
          MedianIncome + Population + PctUnder18 +  
          PctOver65 + PctPoverty + PctUninsured +  
          PctSomeCollege + PctUnemployed,  
          data=training.set)
```


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          PctSomeCollege + PctUnemployed,  
          data=training.set)
```

The RMSE on the training set is just the square root of the mean of the squared residuals:

```
sqrt(mean(residuals(full)^2))  
[1] 1.430723
```

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```

But training set RMSE is “cheating” — let's calculate the RMSE from the test set to get a fairer estimate of performance on new data:

```
sqrt(mean((predict(full, test.set) -  
            test.set$PhysiciansPer10000)^2))  
  
[1] 1.962675
```

Now let's compare to the “reduced” model that backward stepwise regression found:

```
reduced <- lm(PhysiciansPer10000 ~ PctRural + PctUnder18 +  
             PctOver65 + PctSomeCollege + PctUnemployed,  
             data=training.set)
```

RMSE on the training set:

```
sqrt(mean(residuals(reduced)^2))  
  
[1] 1.48333
```

RMSE on the test set—still a little worse:

```
sqrt(mean((predict(reduced, test.set) -  
            test.set$PhysiciansPer10000)^2))  
  
[1] 1.813635
```



Comparing RMSE of the full vs reduced models

The reduced model (with only 5 predictors) does a little *worse* on the training set than the full model (with all 10 predictors), but does much *better* on the test set!

	Training set	Test set
Full model (10 predictors)	1.43	1.96
Reduced model (5 predictors)	1.48	1.81

This suggests that the full model is overfitting the training data, and the reduced model would perform better on new data.



Putting things together

- Use a model selection procedure to help you find the model with the best Adjusted R^2 , AIC, or (for logistic regression) accuracy.
- Better yet, use multiple methods, and it's OK to adjust a model manually: consider practical significance; don't just focus on p -values.
- Consider transforming variables and/or adding interactions to improve the fit and capture key relationships.
- Find the **parsimonious** middle ground between an overly-simple model and extraneous variables—beware of overfitting!
- Use a training/test set to get a direct estimate of model performance on new data.